

(b) Find the number of units and the price per unit that yield the maximum revenue.

(c) Find the maximum revenue.

SOLUTION

(a) Since $S^2 = 12,000 - 2x$ and S is positive, we see that the demand function p is given by

$$S = p(x) = \sqrt{12,000 - 2x}.$$

The domain of p consists of every x such that $12,000 - 2x > 0$, or, equivalently, $2x < 12,000$. Thus, $0 \leq x < 6000$. The graph of p is sketched in Figure 4.57. In theory, there are no sales if the selling price is $\sqrt{12,000}$, or approximately \$109.54, and when the selling price is close to \$0 the demand is close to 6000.

The marginal demand function p' is given by

$$p'(x) = \frac{-1}{\sqrt{12,000 - 2x}}.$$

The negative sign indicates that a decrease in price is associated with an increase in demand.

The revenue function R is given by

$$R(x) = xp(x) = x\sqrt{12,000 - 2x}.$$

Differentiating and simplifying gives us the marginal revenue function R' :

$$R'(x) = \frac{12,000 - 3x}{\sqrt{12,000 - 2x}}$$

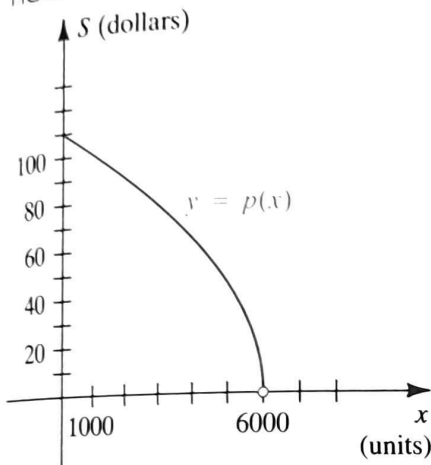
(b) A critical number for the revenue function R is $x = 12,000/3 = 4000$. Since $R'(x)$ is positive if $0 \leq x < 4000$ and negative if $4000 < x < 6000$, the maximum revenue occurs when 4000 units are produced and sold. This corresponds to a selling price per unit of

$$p(4000) = \sqrt{12,000 - 2(4000)} \approx \$63.25.$$

(c) The maximum revenue, obtained from selling 4000 units at \$63.25 per unit, is

$$4000(63.25) = \$253,000.$$

FIGURE 4.57



EXERCISES 4.7

Exer. 1–8: A point moving on a coordinate line has position function s . Find the velocity and acceleration at time t , and describe the motion of the point during the indicated time interval. Illustrate the motion by means of a diagram of the type shown in Figure 4.54.

1 $s(t) = 3t^2 - 12t + 1$; $[0, 5]$

2 $s(t) = t^2 + 3t - 6$; $[-2, 2]$

3 $s(t) = t^3 - 9t + 1$; $[-3, 3]$

4 $s(t) = 24 + 6t - t^3$; $[-2, 3]$

5 $s(t) = -2t^3 + 15t^2 - 24t - 6$; $[0, 5]$

6 $s(t) = 2t^3 - 12t^2 + 6$; $[-1, 6]$

7 $s(t) = 2t^4 - 6t^2$; $[-2, 2]$

8 $s(t) = 2t^3 - 6t^5$; $[-1, 1]$

Exer. 9–10: An automobile rolls down an incline, traveling $s(t)$ feet in t seconds. [a] Find its velocity at $t = 3$. [b] After how many seconds will the velocity be k ft/sec?

9 $s(t) = 5t^2 + 2$; $k = 28$

10 $s(t) = 3t^2 + 7$; $k = 88$

Exer. 11–12: A projectile is fired directly upward with an initial velocity of v_0 ft/sec, and its height (in feet) above the ground after t seconds is given by $s(t)$. Find [a] the velocity and acceleration after t seconds, [b] the maximum height, and [c] the duration of the flight.

11 $v_0 = 144$: $s(t) = 144t - 16t^2$

12 $v_0 = 192$: $s(t) = 100 + 192t - 16t^2$

Exer. 13–16: A particle in simple harmonic motion has position function s , and t is the time in seconds. Find the amplitude, period, and frequency.

13 $s(t) = 5 \cos \frac{\pi}{4} t$

14 $s(t) = 4 \sin \pi t$

15 $s(t) = 6 \sin \frac{2\pi}{3} t$

16 $s(t) = 3 \cos 2t$

17 The electromotive force V and current I in an alternating-current circuit are given by

$$V = 220 \sin 360\pi t$$

$$I = 20 \sin \left(360\pi t - \frac{\pi}{4} \right).$$

Find the rates of change of V and I with respect to time at $t = 1$.

18 The annual variation in temperature T (in $^{\circ}\text{C}$) in Vancouver, B.C., may be approximated by the formula

$$T = 14.8 \sin \left[\frac{\pi}{6} (t - 3) \right] + 10,$$

where t is in months, with $t = 0$ corresponding to January 1. Approximate the rate at which the temperature is changing at time $t = 3$ (April 1) and at time $t = 10$ (November 1). At what time of the year is the temperature changing most rapidly?

19 The graph in the figure shows the rise and fall of the water level in Boston Harbor during a particular 24-hour period.

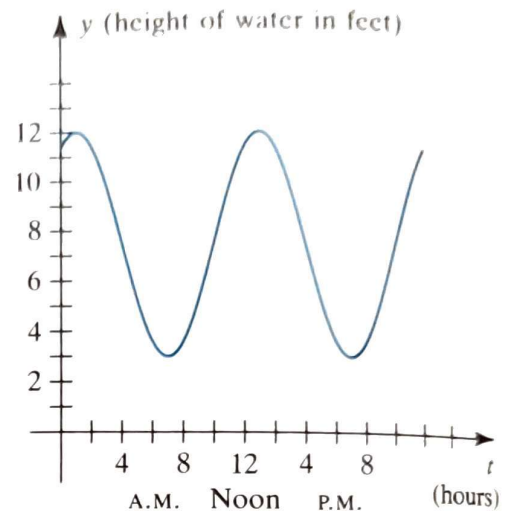
[a] Approximate the water level y by means of an expression of the form

$$y = a \sin (bt + c) + d,$$

with $t = 0$ corresponding to midnight.

[b] Approximately how fast is the water level rising at 12 noon?

EXERCISE 19

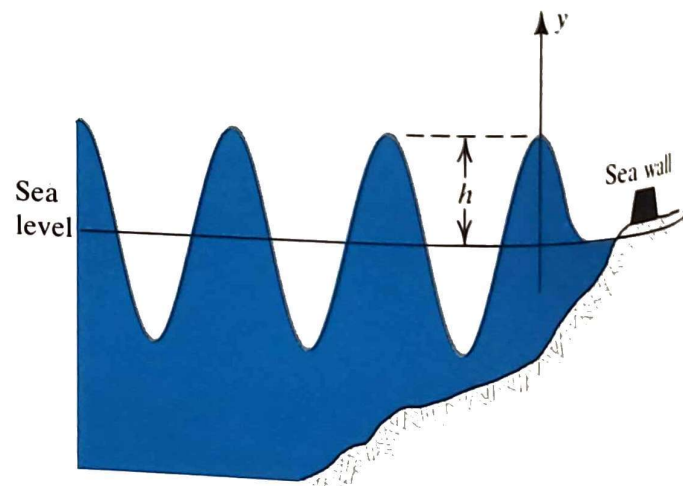


20 A tsunami is a tidal wave caused by an earthquake beneath the sea. These waves can be more than a hundred feet in height and can travel at great speeds. Engineers sometimes represent tsunamis by an equation of the form $y = a \cos bt$. Suppose that a wave has height $h = 25$ ft and period 30 minutes and is traveling at the rate of 180 ft/sec.

[a] Let (x, y) be a point on the wave represented in the figure. Express y as a function of t if $y = 25$ ft when $t = 0$.

[b] How fast is the wave rising (or falling) when $y = 10$ ft?

EXERCISE 20

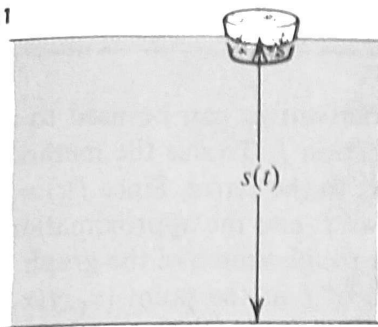


21 A cork bobs up and down in a lake. The distance from the bottom of the lake to the center of the cork at time $t \geq 0$ is given by $s(t) = \cos \pi t + 12$, where $s(t)$ is in inches and t is in seconds. (See figure on page 231.)

[a] Find the velocity of the cork at $t = 0, \frac{1}{2}, 1, \frac{3}{2},$ and 2 .

[b] During what time intervals is the cork rising?

EXERCISE 21



22 A particle in a vibrating spring is moving vertically such that its distance $s(t)$ from a fixed point on the line of vibration is given by $s(t) = 4 + \frac{1}{25} \sin 100\pi t$, where $s(t)$ is in centimeters and t is in seconds.

(a) How long does it take the particle to make one complete vibration?

(b) Find the velocity of the particle at $t = 1, 1.005, 1.01$, and 1.015 .

Exer. 23–24: Show that $s''(t) = -\omega^2 s(t)$.

23 $s(t) = k \cos(\omega t + b)$

24 $s(t) = k \sin(\omega t + b)$

25 A point $P(x, y)$ is moving at a constant rate around the circle $x^2 + y^2 = a^2$. Prove that the projection $Q(x, 0)$ of P onto the x -axis is in simple harmonic motion.

26 If a point P moves on a coordinate line such that

$$s(t) = a \cos \omega t + b \sin \omega t,$$

show that P is in simple harmonic motion by

(a) using the remark following Definition (4.22)

(b) using only trigonometric methods (Hint: Show that $s(t) = A \cos(\omega t - c)$ for some constants A and c .)

Exer. 27–28: A point moving on a coordinate line has position function s . (a) Graph $y = s(t)$ for $0 \leq t \leq 5$. (b) Approximate the point's position, velocity, and acceleration at $t = 0, 1, 2, 3, 4$, and 5 .

27 $s(t) = \frac{10 \sin t}{t^2 + 1}$

28 $s(t) = \frac{5 \tan(\frac{1}{4}t)}{2t + 1}$

Exer. 29–32: If C is the cost function for a particular product, find (a) the cost of producing 100 units and (b) the average and marginal cost functions and their values at $x = 100$.

29 $C(x) = 800 + 0.04x + 0.0002x^2$

30 $C(x) = 6400 + 6.5x + 0.003x^2$

31 $C(x) = 250 + 100x + 0.001x^3$

32 $C(x) = 200 + 0.01x + \frac{100}{x}$

33 A manufacturer of small motors estimates that the cost (in dollars) of producing x motors per day is given by $C(x) = 100 + 50x + (100/x)$. Compare the marginal cost of producing five motors with the cost of producing the sixth motor.

34 A company conducts a pilot test for production of a new industrial solvent and finds that the cost of producing x liters of each pilot run is given by the formula $C(x) = 3 + x + (10/x)$. Compare the marginal cost of producing 10 liters with the cost of producing the 11th liter.

Exer. 35–36: For the given demand and cost functions, find (a) the marginal demand function, (b) the revenue function, (c) the profit function, (d) the marginal profit function, (e) the maximum profit, and (f) the marginal cost when the demand is 10 units.

35 $p(x) = 50 - 0.1x$; $C(x) = 10 + 2x$

36 $p(x) = 80 - \sqrt{x-1}$; $C(x) = 75x + 2\sqrt{x-1}$

37 A travel agency estimates that, in order to sell x package-deal vacations, it must charge a price per vacation of $1800 - 2x$ dollars for $1 \leq x \leq 100$. If the cost to the agency for x vacations is $1000 + x + 0.01x^2$ dollars, find

(a) the revenue function

(b) the profit function

(c) the number of vacations that will maximize the profit

(d) the maximum profit

38 A manufacturer determines that x units of a product will be sold if the selling price is $400 - 0.05x$ dollars for each unit. If the production cost for x units is $500 + 10x$, find

(a) the revenue function

(b) the profit function

(c) the number of units that will maximize the profit

(d) the price per unit when the marginal revenue is 300

39 A kitchen specialty company determines that the cost of manufacturing and packaging x pepper mills per day is $500 + 0.02x + 0.001x^2$. If each mill is sold for \$8.00, find

(a) the rate of production that will maximize the profit

(b) the maximum daily profit

40 A company that conducts bus tours found that when the price was \$9.00 per person, the average number of customers was 1000 per week. When the company reduced the price to \$7.00 per person, the average number of customers increased to 1500 per week. Assuming that the demand function is linear, what price should be charged to obtain the greatest weekly revenue?